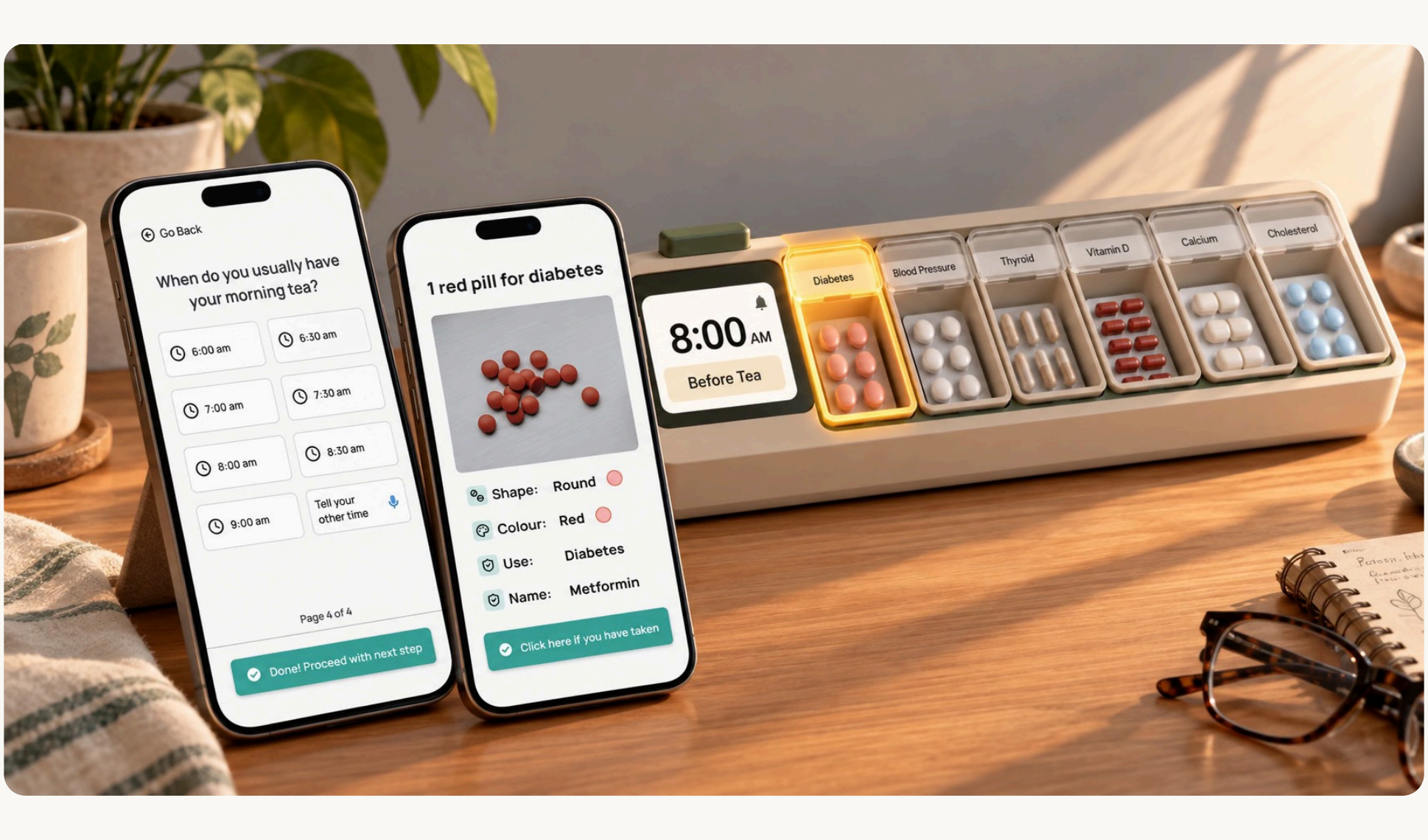
Simplifying Polypharmacy for Older Adults

Designing an accessible experience for managing multiple medications and everyday care.

My Role
Ux Research, Use interviews, Behavioural Research, Interaction design, Information architecture, Visual design.

Project Type
UX Research, Service design, application design, Product design

Tools Used


8:00 AM
Before Tea
1 red pill for diabetes
When do you usually have your morning tea?

What does the current scenario look like?

1 **Managing around 4 to 12 medicines each day**
Managing care can mean multiple medicines, frequent doctor visits, and changing prescriptions.

2 **Medical Information Stored Separately**
Medicine details and medical records are often scattered, making quick access difficult.

3 **Multiple Doctor's visit and Prescription change**
Frequent doctor visits lead to changing prescriptions and make care harder to manage.

4 **Tedious Pill Arrangement**
Sorting medicines into pill boxes by day and time is repetitive and time-consuming.

Qualitative Research

Interviewed 7elderly people, 9 family members, 1 medical student

"When my routine changes, I sometimes forget to take them, which affects my health."

"I remember my medicines through routines, like after brushing or before breakfast."

"I do not want to depend on my family all the time"

" Medical records are scattered"

"Sometimes we need a Caregiver, but it is hard to find them"

"I want to use digital tools, but they're often difficult to understand and use."

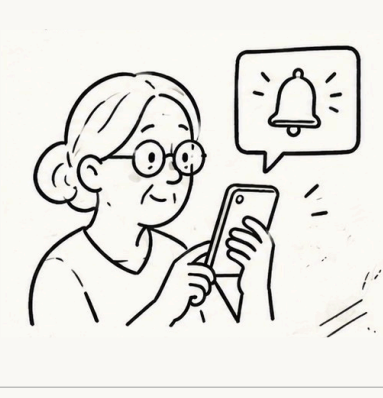
Quantitative Research

Surveyed 102 people

70%
"Still rely on memory"

65%
"Store medical records phscially"

52%
"See value in medicine remainders"



Secondary Research

Medicines are remembered by how they look
They often recognise medicines by their shape, size, colour or where they keep it.

Medicines are connected to routines
They often remember medicines as before tea, after lunch...

Remainders helpful when the medicines are new
Helpful until they get used to it

A digital solution needs to be genuinely useful
If their existing solution is working for them elready, they want something more from a digital solution

Elderly, wanted more features
Like medication schedules, reminders, dose tracking, refill alerts, and medicine information in one place.

Problem Statement

"How might we simplify medication and everyday care while keeping families involved?"

User personas

01 ELDERLY PERSON

Meena Joshi · 68
Retired teacher · Lives with her husband

Meena takes multiple medicines every day. She has developed her own way of remembering them through their appearance and her daily routine.

Needs
- Easily recognise the right medicine
- Store medical history and records
- Add medicines and records easily
- Get reminders when needed
- Adaptable remainder for routine change

Frustrations
- Similar-looking medicines get confusing
- Changing prescriptions disturb her routine
- Too many steps or information can feel overwhelming.

02 FAMILY CAREGIVER

Rahul Joshi · 39
Working professional · Meena's son

Rahul helps manage her mothers healthcare, but work means he cannot always be around to check medicines, appointments or accompany her

Needs
- Know whether medicines are taken
- Quickly add or Update medicines
- Keep records and appointments together
- Find trustable caregiver
- Adaptable remainder for routine change

Frustrations
- Managing many medicines takes time
- Medical information is scattered
- Hard to know what happened when he wasn't there

The solution

Two connected digital products
one designed for elderly users to manage their medicines, and another for family caregivers to support and manage their care.

Grouped medicine system
Medicines taken around the same routine are grouped togeth

More personal routine remainders
Remainders can be set around smaller everyday routines like before tea, after tea...

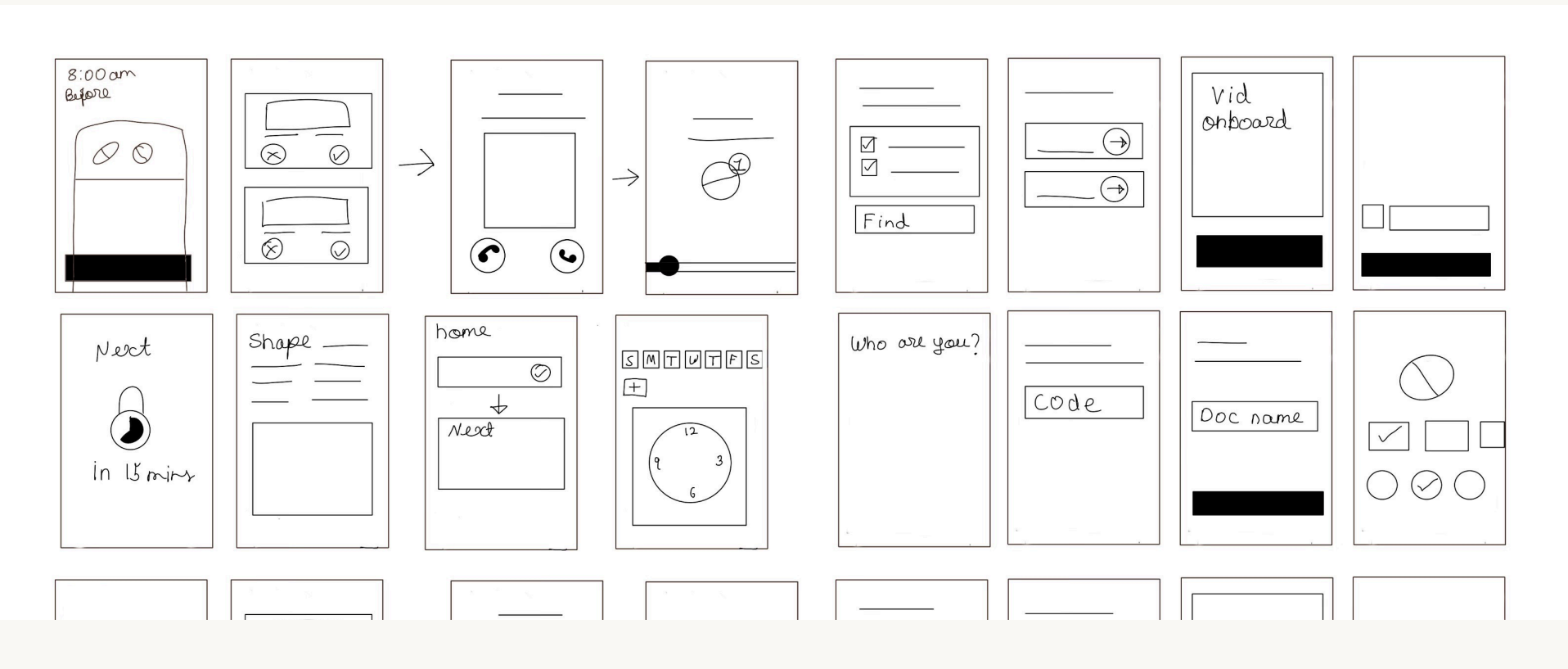
Easy to recognise medicine remainders
Remainders that show the medicine's image, colour, shape and familiar name

Easy medicine management
Medicines can be added quickly, updated, or turned off when no longer needed

Family medicine tracking
Family members can see which medicines were taken or missed and keep track of the medicines

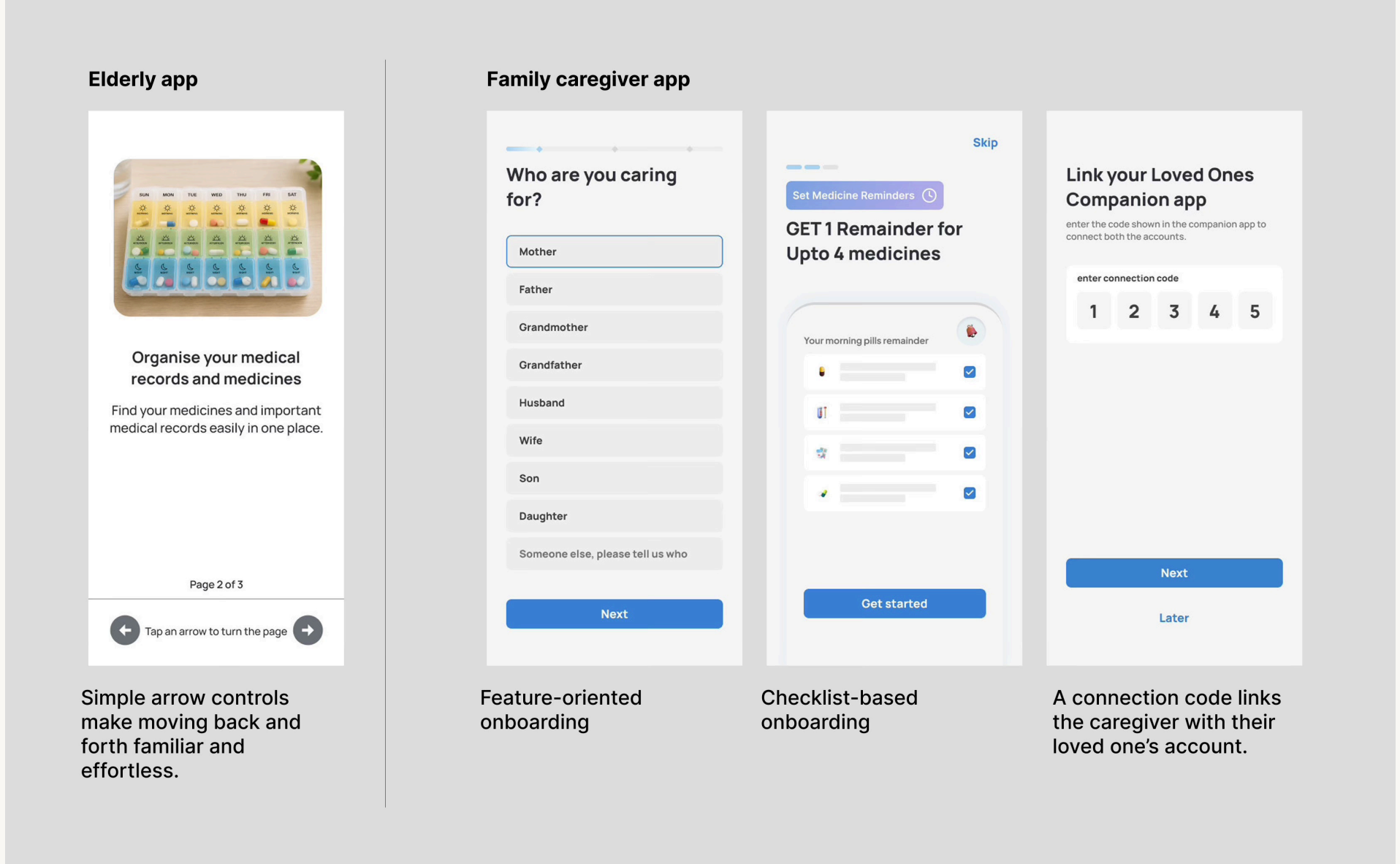
Book caregiver for specific needs
Families can find caregiver for specific needs and timeline

Wireframes


8:00am
Before
Next
In 15 mins
Shape
home
Next
Find
Who are you?
Code
Vid onboard
Doc name

High-fidelity screens

Onboarding


Elderly app
Family caregiver app
Organise your medical records and medicines
Who are you caring for?
GET 1 Remainder for Upto 4 medicines
Link your Loved Ones Companion app

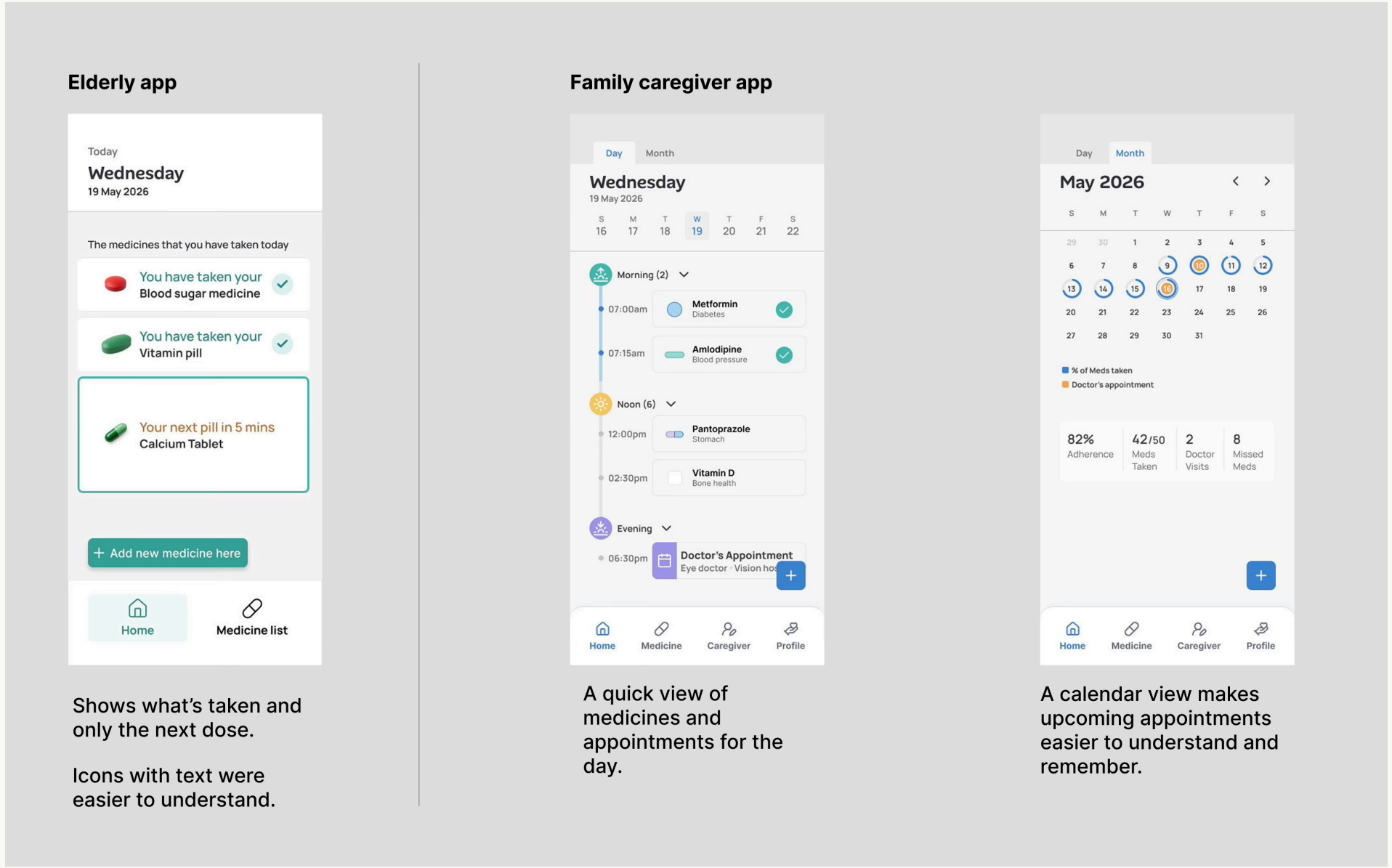
Simple arrow controls make moving back and forth familiar and effortless.

Feature-oriented onboarding

Checklist-based onboarding

A connection code links the caregiver with their loved one's account.

Home screen


Elderly app
Family caregiver app
Wednesday
May 2026

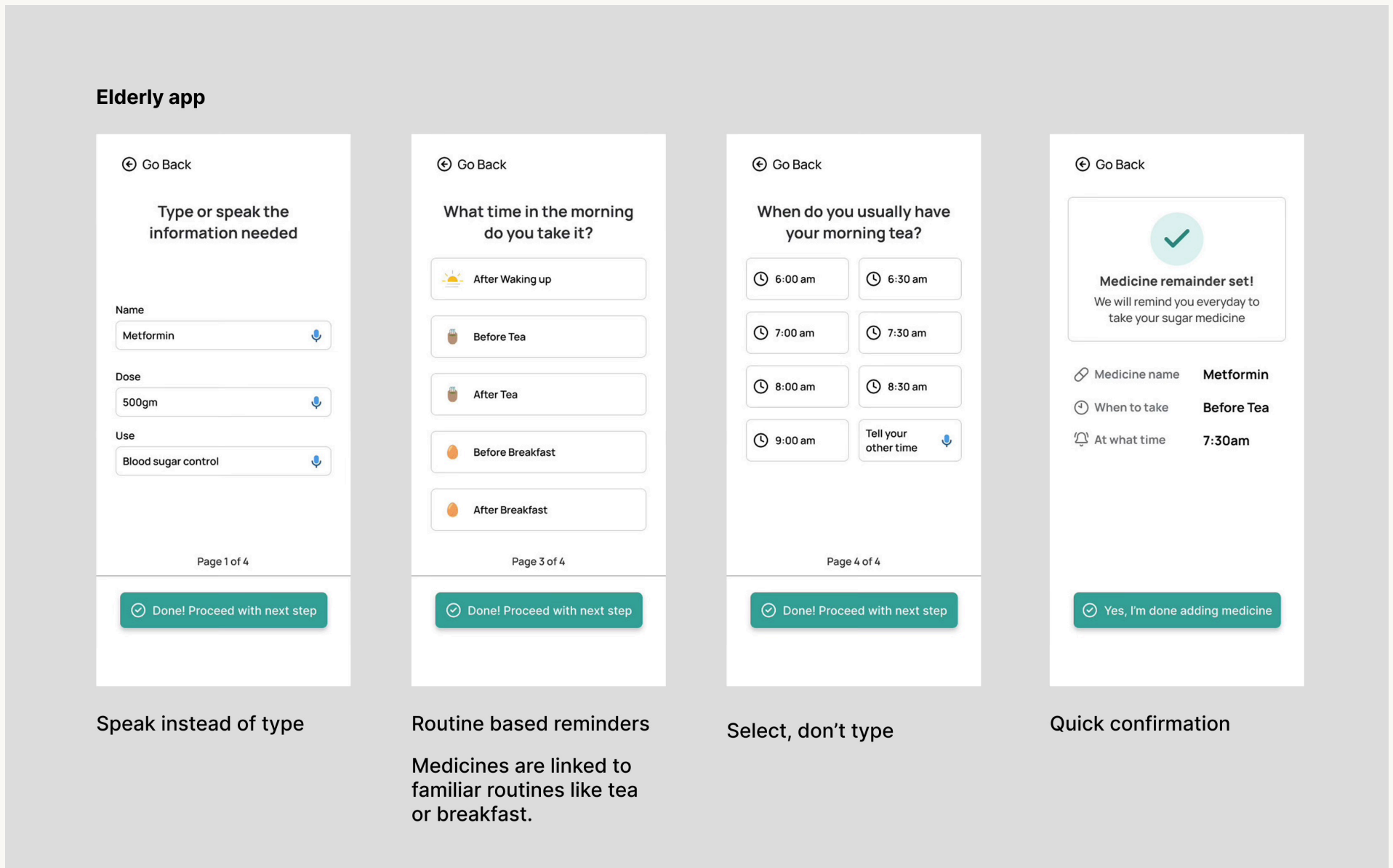
Shows what's taken and only the next dose.

Icons with text were easier to understand.

A quick view of medicines and appointments for the day.

A calendar view makes upcoming appointments easier to understand and remember.

Adding medicine - Elderly app


Elderly app
Type or speak the information needed
What time in the morning do you take it?
When do you usually have your morning tea?
Medicine remainder set!

Speak instead of type

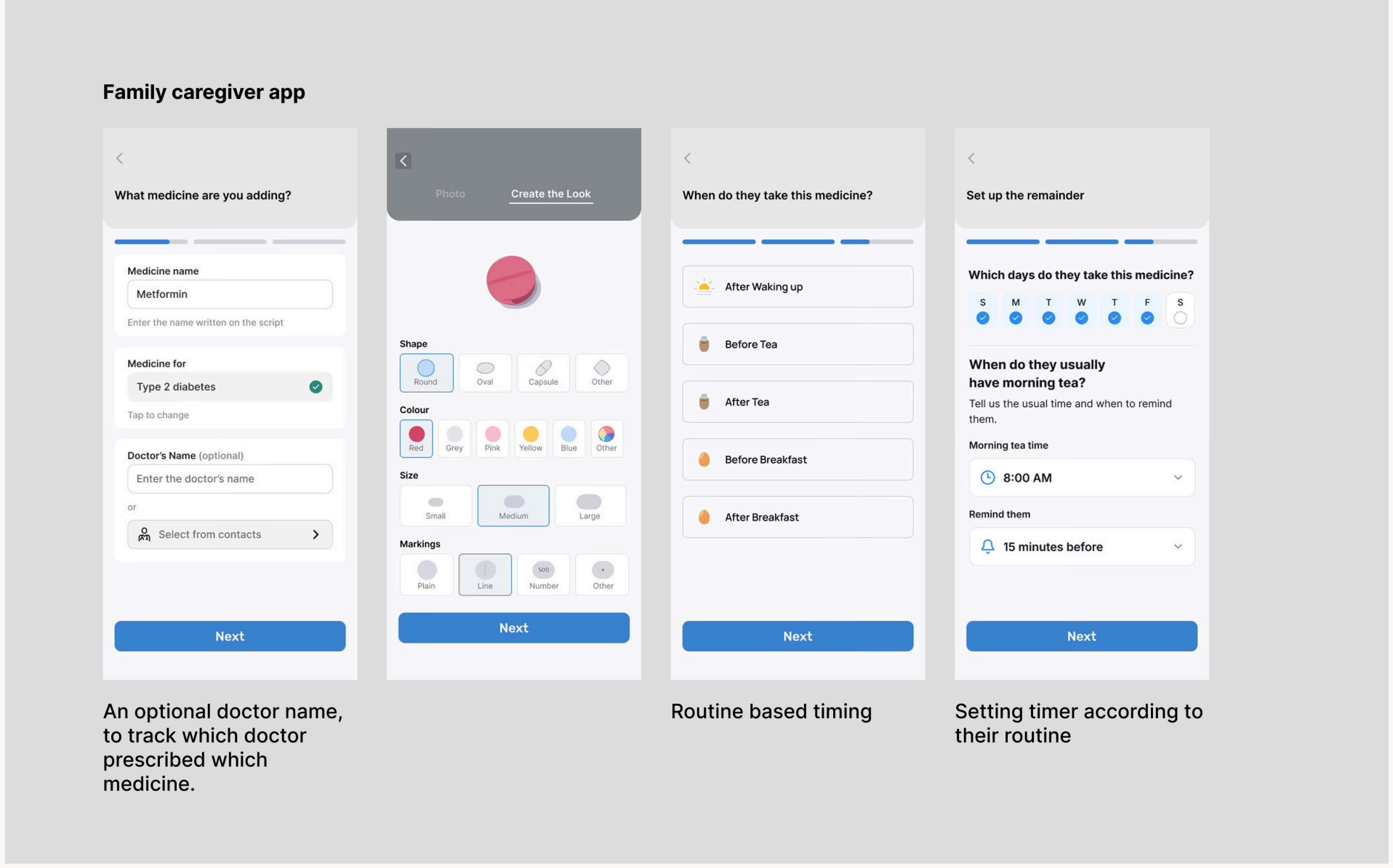
Routine based reminders

Medicines are linked to familiar routines like tea or breakfast.

Select, don't type

Quick confirmation

Adding medicine - Family caregiver app

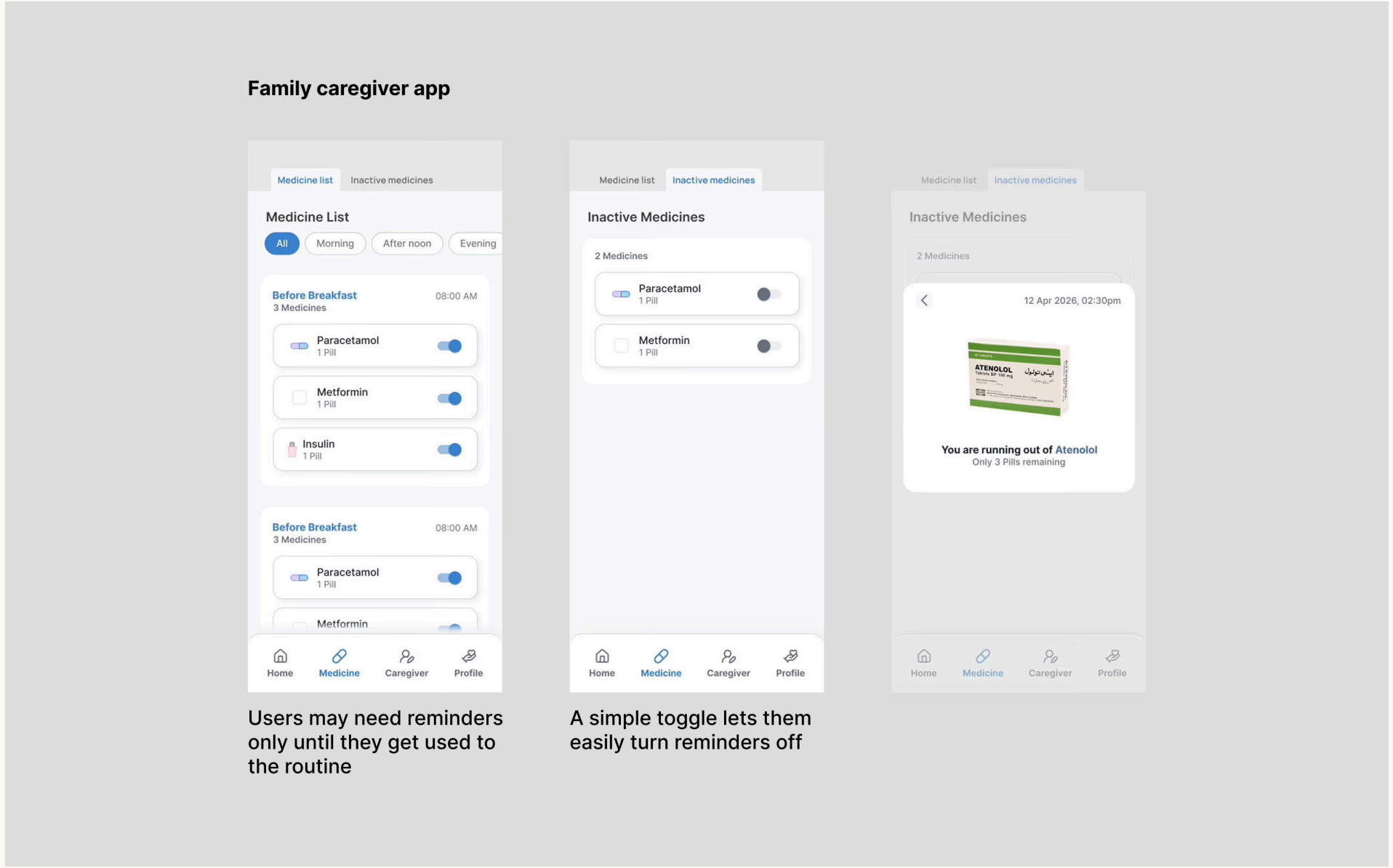

Family caregiver app
What medicine are you adding?
Create the Look
When do they take this medicine?
Set up the remainder

An optional doctor name, to track which doctor prescribed which medicine.

Routine based timing

Setting timer according to their routine

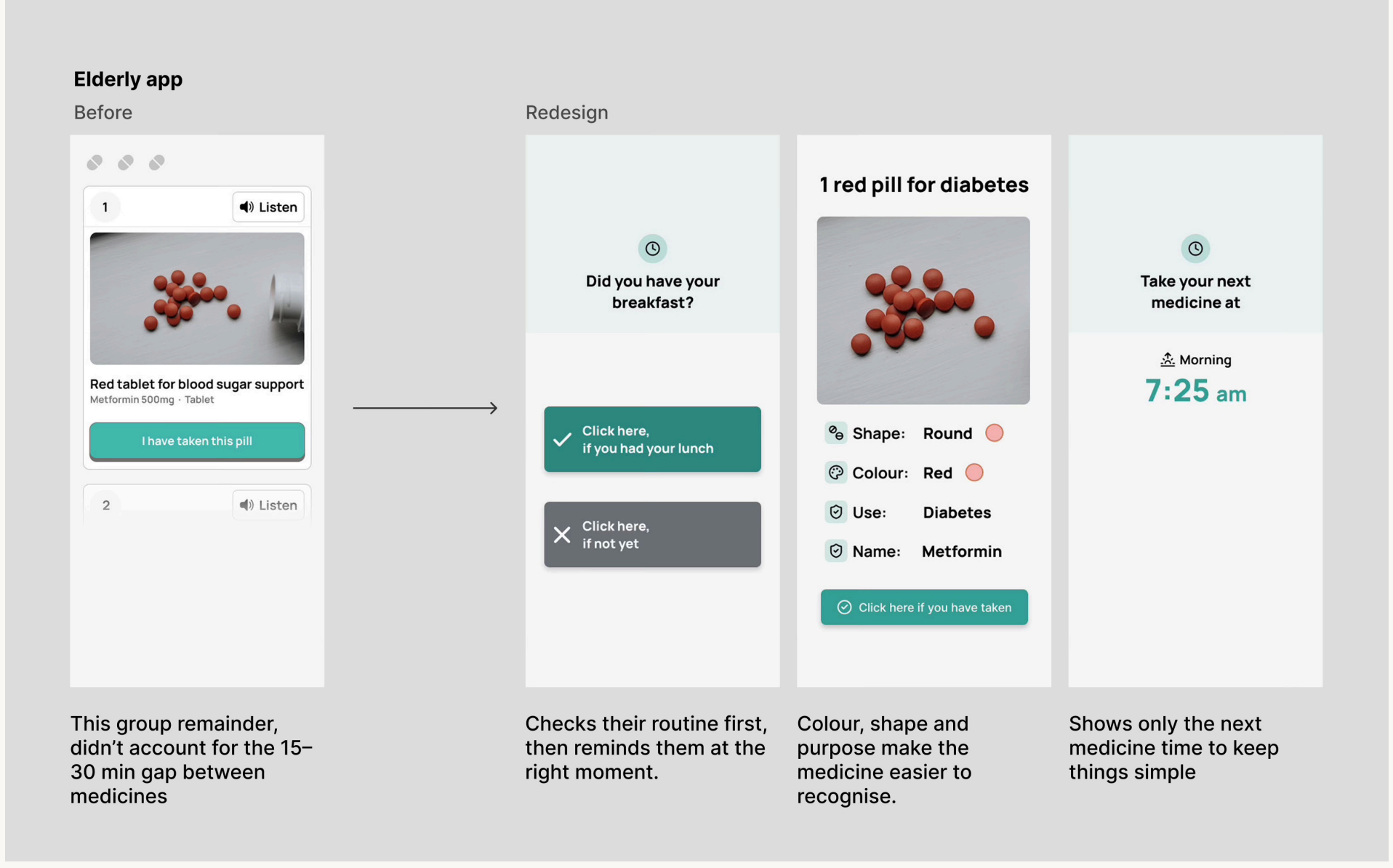
Managing medicine


Family caregiver app
Medicine List
Inactive Medicines

Users may need reminders only until they get used to the routine

A simple toggle lets them easily turn reminders off

Medicine remainders


Elderly app
Before
Redesign
Did you have your breakfast?
1 red pill for diabetes
Take your next medicine at
7:25 am

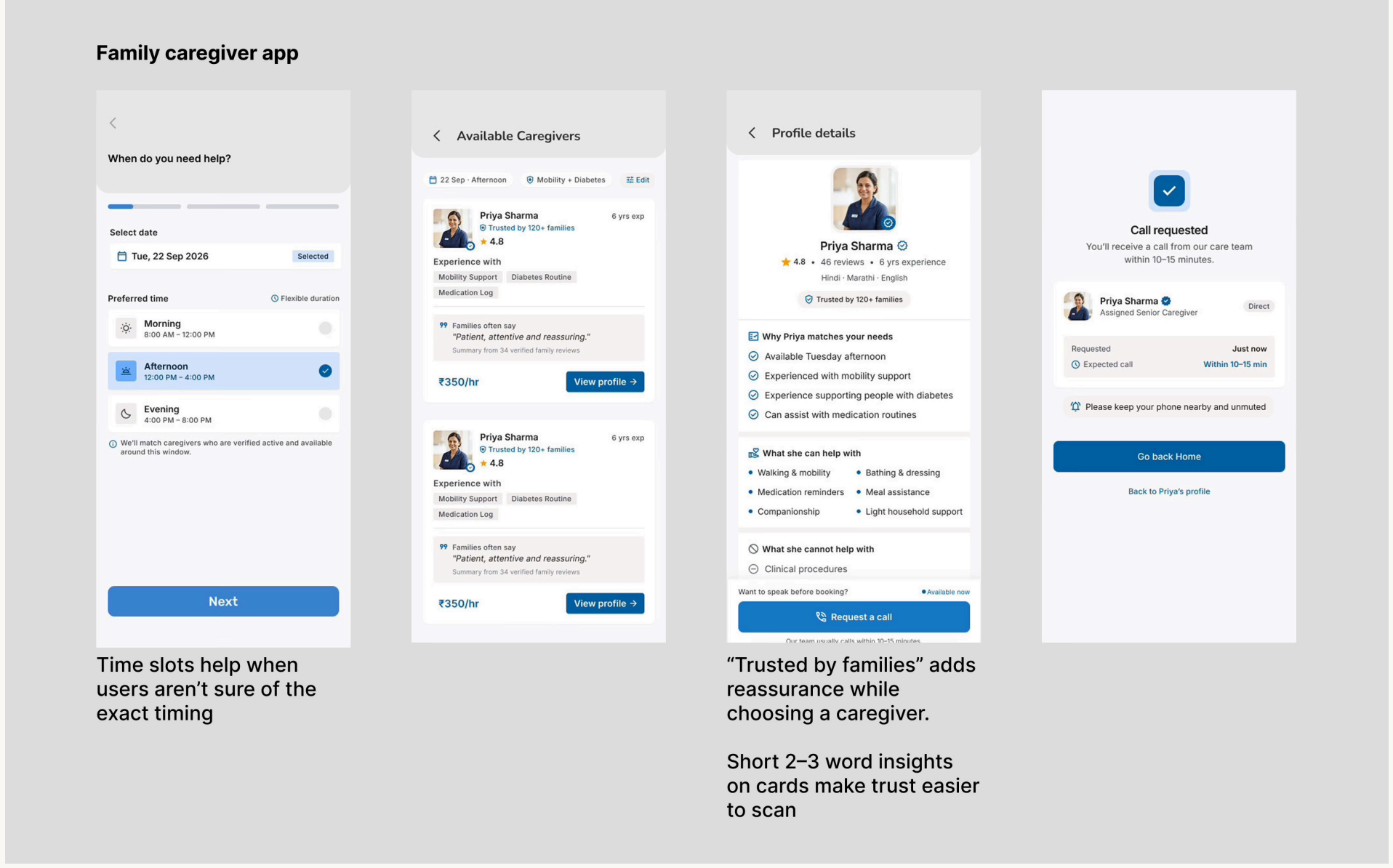
This group remainder, didn't account for the 15–30 min gap between medicines

Checks their routine first, then reminds them at the right moment.

Colour, shape and purpose make the medicine easier to recognise.

Shows only the next medicine time to keep things simple

Caregiver booking


Family caregiver app
When do you need help?
Available Caregivers
Profile details
Call requested

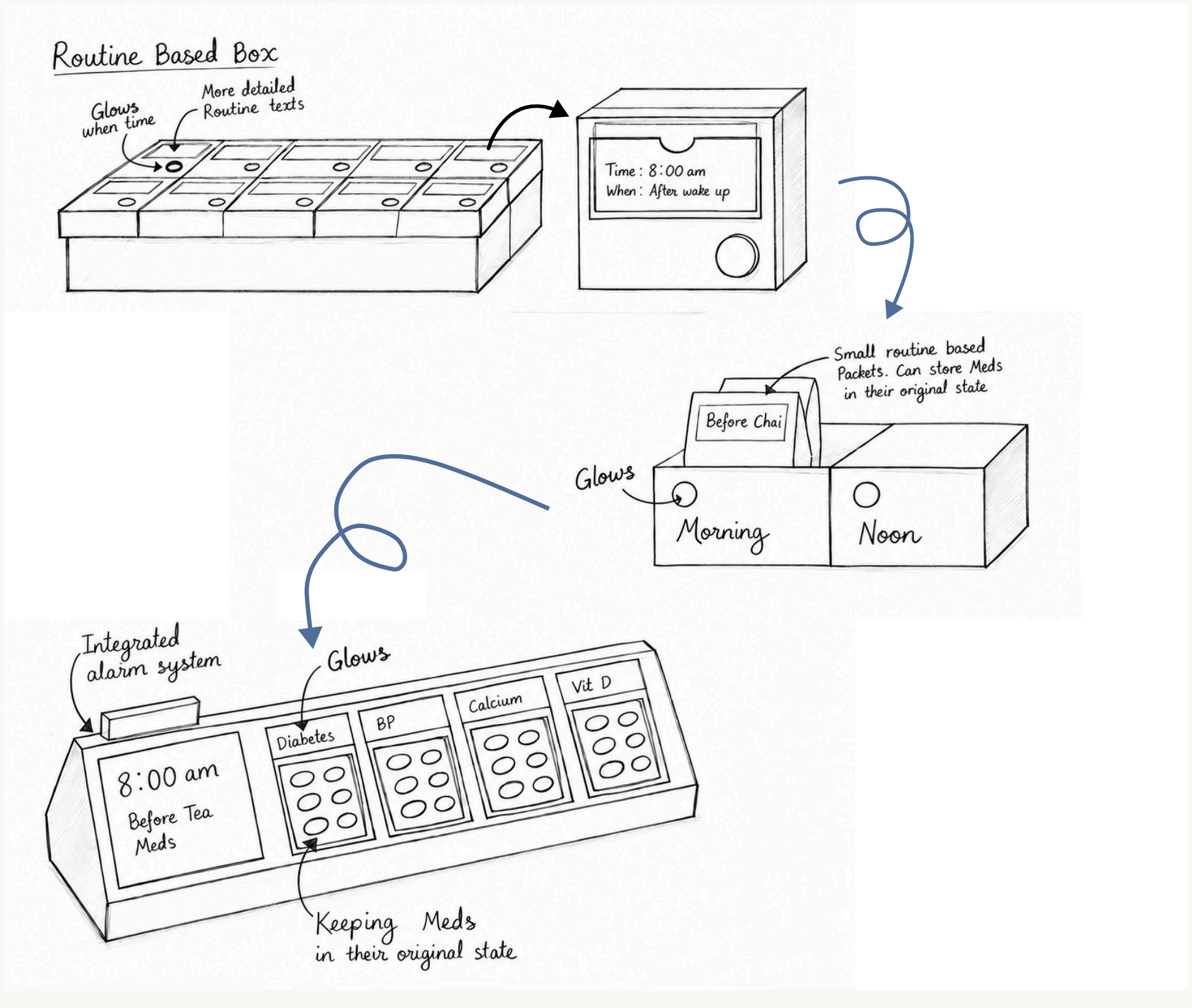
Time slots help when users aren't sure of the exact timing

"Trusted by families" adds reassurance while choosing a caregiver.

Short 2–3 word insights on cards make trust easier to scan

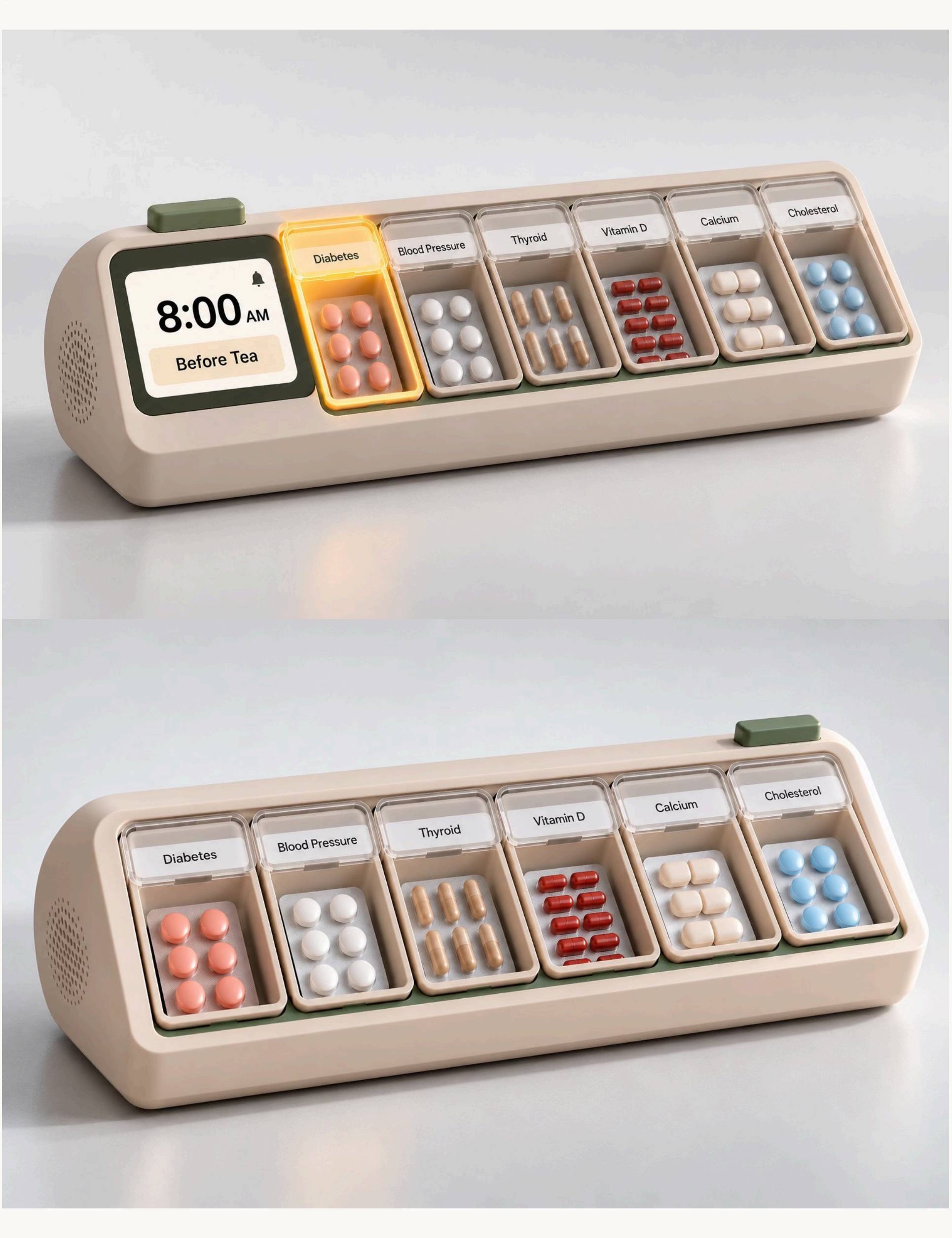
Pill box design

Pain point
users find it tedious to manually sort multiple medicines into day and night time based compartments, a repetitive task they have redo every week.


Routine Based Box
Glows when time
More detailed Routine texts
Time : 8:00 am
When : After wake up
Small routine based Packets. Can store Meds in their original state
Before Chai
Glows
Morning
Noon
Integrated alarm system
Glows
Diabetes
BP
Calcium
Vit D
8:00 am
Before Tea Meds
Keeping Meds in their original state

Pill box final mockup

Solution
Keep medicines in their original state. when it's time to take one, the corresponding section glows and the alarm rings, guiding the user to the right medicine.


8:00 AM
Before Tea
Diabetes
Blood Pressure
Thyroid
Vitamin D
Calcium
Cholesterol

Elderly app usability testing

I tested the key flows to understand whether users could complete everyday medication tasks easily and without confusion.

12 Participants 50+ yrs,

4 Key flows tested

High-fidelity prototype

Moderated testing
Task-based + post-task interview

#	FLOW TESTED	AVG. TIME TAKEN	TASK COMPLETION	ERROR RATE	KEY ERRORS OBSERVED	KEY INSIGHT
1	Add a medicine	~50 sec	~97%	3%	Small input fields; users were unsure when to move to the next input	One question per screen could make the flow clearer
2	Recognise reminder	~10 sec	~92%	~8%	Medicines were easy to recognise, but users took time to read button labels	Keep actions shorter and easier to scan
3	Mark medicine as taken	~15 sec	75%	25%	3/12 didn't understand that they had to tap the button to mark it taken	The "Taken" action needs a clearer visual cue
4	Check today's medicines	<10 sec	100%	0%	No errors observed	Day-wise medicine status was immediately understood

100%
understood today's medicine list

~50 sec
to add a medicine

~10 sec
to recognise a reminder

75%
understood the "mark as taken" interaction button without confusion

Familycaregiver app usability testing

I tested the key flows to understand how easily family caregivers could manage medicines, track adherence and find care when needed.

22 Participants

4 Key flows tested

High-fidelity prototype

Moderated testing
Task-based + post-task interview

#	FLOW TESTED	AVG. TIME TAKEN	TASK COMPLETION	ERROR RATE	KEY ERRORS OBSERVED	KEY INSIGHT
1	Add a medicine	~50 sec	100%	0%	Users thought the auto-detected "medicine purpose" needed manual input; "Create the look" was unclear by name	Auto-detection needs to be more visible, and "Create the look" needs clearer wording
2	Turn off medicine	~5 sec	100%	0%	No major errors observed	Users could quickly understand and control medicine reminders
3	Track medication	10–20 sec	91%	9%	2/22 couldn't clearly differentiate between taken and upcoming medicines; Clinical names were difficult to understand	Highlight familiar labels like "Diabetes medicine" instead of clinical names
4	Book a caregiver	2–5 min	100%	0%	No major usability errors observed	Caregiver profiles gave users assurance before booking

100%
completed the Add Medicine flow

~5 sec
to turn off medicine reminders

91%
understood medication status

100%
completed the caregiver booking flow

Outcomes

The simplicity worked
Both elderly users and family caregivers found the core flows easy to understand and use.

Medicine history reduced uncertainty
Users could quickly check if they had already taken a medicine that day.

Users wanted more for continued use
8+/12 users wanted easier record adding and other features like info of meds, drug interaction, and many other features

Medicine status on family home screen needed more clarity
Caregivers struggled with clinical medicine names, and 2 users missed the ✓ taken status.

Caregiver profiles built trust
Profiles increased assurance, but waiting for a callback didn't work for emergencies.

CASE CLOSED.